Where we are on the course

We go at a pace that makes sense and adjust the depth/breadth of topics

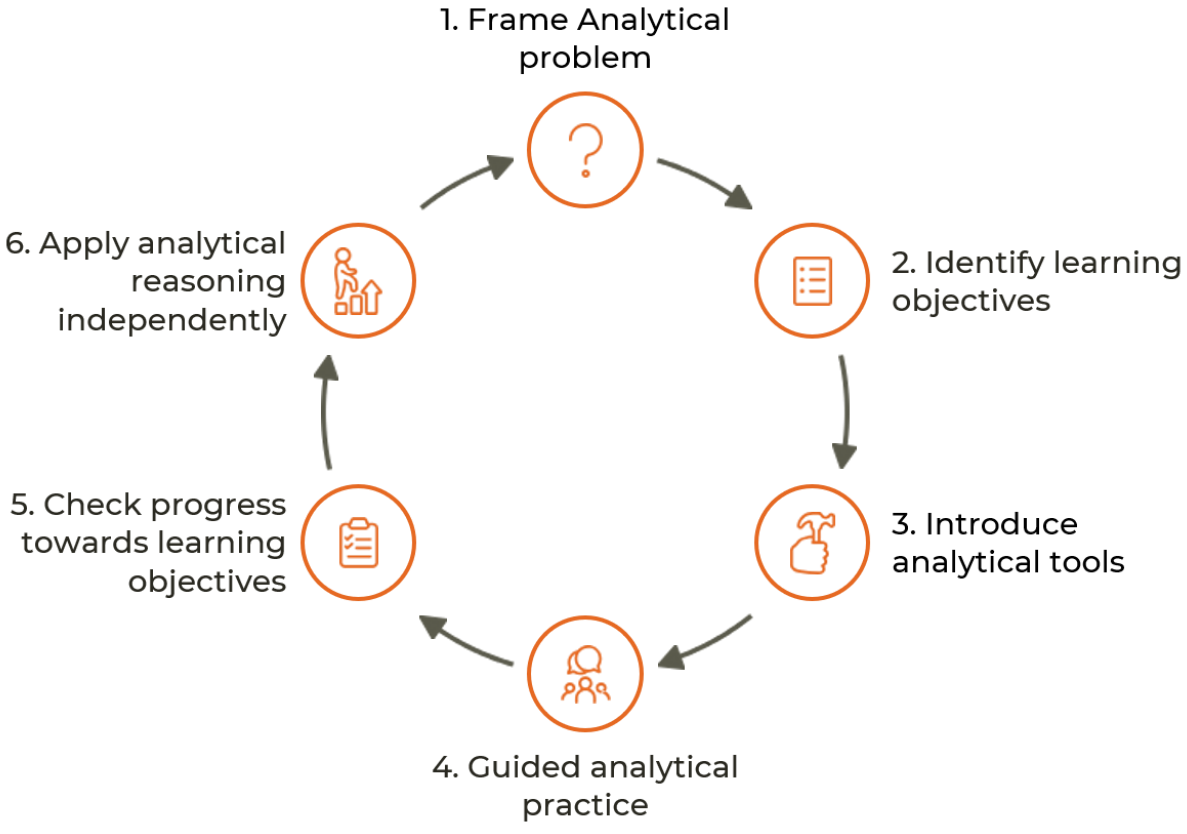
Period I

- **L1: Course orientation and foundations of marine data analysis**
- L2: Exploratory Data Analysis (EDA)
- L3 & L4: Data pre-processing
- L5: From engineering questions to analytical models
- L6: Marine data analysis in practice

Period II

- L7: Regression modelling, evaluation, and refinement
- L8 & L9: AIS data analytics
- L10 & L11: Cluster analysis
- L12: AI-assisted qualitative data analysis
- Project demonstration

Problem based learning cycle



Aalto-yliopisto
Insinööritieteiden
korkeakoulu

Introduction to Marine Data

MEC-E2016 Marine Data Analysis

Reference: Contents of this lecture are partially covered in

1. Data mining: Concepts and techniques. Chapter 2.
2. Guide to Maritime Informatics. Chapter 2.

Assignment 1

Assume that you have secured an internship at Helsinki Shipyard and joined its research team. Your first task is to model propulsion shaft power using relevant variables available in the Cruiseships database. Before beginning the modelling, you need to determine whether the data are ready and which variables may be useful. Use exploratory data analysis (EDA) to investigate the following:

Grade 1 to 2: Data quality: Identify up to three important data-quality issues, such as missing values, unsuitable data types, inconsistent categories, or potentially unusual values. Explain why each may require attention.

Grade 3 to 4: Variation in the variables: Examine how the variables are distributed across the dataset. Explain the main patterns you observe, including their typical values, variability, and any notable skewness, imbalance, or unusual observations.

Grade 5: Variables relevant to propulsion shaft power: Investigate how propulsion shaft power relates to the other variables in the dataset. Also examine whether some variables are closely related to each other and may therefore provide similar information. Based on these relationships, identify which variables should be considered when modelling shaft power and explain your reasoning.

Present and interpret your findings in the report, supported by appropriate numerical summaries and/or visualizations. Use a Jupyter Notebook or another appropriate tool of your choice to perform the analysis.

Learning objectives

At the end of this lecture, you should be able to

- Describe different types of marine data
- Understand the motivation behind marine data analysis
- Define model, variable and function
- Use your programming environment – Jupyter notebook
- Demonstrate how to import and export data in Python

Why Data Analysis?

- We live in a world where vast amounts of data are collected daily (or even per hour or per minute or per second?)

- Data analysis give you the means to discover knowledge from the data – Think of the google queries you perform each day

Marine Data

- Vessel related data
  - These data sources offer information relating to the life-cycle of every ship, from the shipbuilding phase to its current state, including its seaworthiness or unseaworthiness, the history of its ownership, its structural, mechanical and operational characteristics and all the changes made to the ship overtime, the ship's more recent trajectories and voyages around the globe as well as its current position at sea or at anchor.

- Sea related data
  - Sea-related data include data that describe the area of about the 70% of the globe that is covered by sea water. The data describe this area from the points of view of both above and below the surface of the water and include data for bathymetry, undersea cables, protected areas, wetlands, fish population, human-made structures, etc.

Marine Data

- Land related data
  - This section presents data that are related to the area amounting to about 30% of the globe which is land, but which is in some way also related to its aquatic area. For example, the World Port Index lists about 4,000 ports across the entire globe, describing their characteristics and the known facilities and available services which they provide.

- Map and imagery related data
  - Electronic nautical chart, radar images, satellite images

- Climate and Sea conservancy related data
  - Weather conditions, the climate system (climate is long-term average of the weather), the impact of human activity on the health of the ocean

- Can you think of anything else?

There are several marine database examples on the course page.

Marine Data Analysis – Case study

- You are a marine data analyst at a regional maritime surveillance center responsible for monitoring vessel activity in protected waters.
- A commercial vessel enters a strictly protected marine reserve. Its speed drops from approximately 11 knots to between 3 and 4 knots.
- It remains inside the reserve for more than four hours before returning to 11 knots and continuing its journey.
- A simple analysis identifies only a period of unusually slow movement. Would you raise an alert?

Marine Data Analysis – Case study

- Additional information
  - The vessel is registered as a bottom trawler—a fishing vessel that tows a net along or close to the seabed.
  - Inside the reserve, it follows several long, approximately parallel tracks and repeatedly turns back in the opposite direction.

- Does these added information change your interpretation?

Marine Data Analysis – Case study

- Domain knowledge
  - Trawlers commonly travel at steady speeds of approximately 2.5–5.5 knots while towing their nets.
  - Bottom trawling is prohibited inside the reserve.

- Conclusion
  - The speed alone indicates only slow movement.
  - When combined with the vessel type, movement pattern and location, it becomes evidence of **possible** prohibited fishing.
  - Further evidence would still be needed to confirm it.

Why Marine Data Analysis?

- Application of these marine data results in challenging use-case scenarios
  - discovering valuable behavioral patterns of moving objects
  - identifying different types of activities in a region of interest
  - estimating fishing pressure or environmental fingerprint etc.

- Transforming the raw data into an appropriate form for the application of data analytic tasks is not straightforward and need domain knowledge
  - noise elimination, compression and trajectory identification

- After transformation, you can apply usual techniques (classification, regression, clustering) but both setting inputs and interpreting outputs need domain knowledge

Terminologies

- Model: A mathematical model consists of variables and functions

- Variables represent aspects of the data and the model
  - A numerical variable can contain the ship speed
  - A 3-dimensional variable may represent seawater temperature across latitude, longitude, and depth.
  - Variables can also represent abstract aspects like cruise ship passenger satisfaction.

  - The nouns attribute, dimension, feature, and variable are often used interchangeably

- Functions relate the variables to each other
  - Ship speed may be expressed as a function of *propulsion power, loading condition, and environmental conditions*

Modeling – variables of interest

- **The purpose of the model – how to use it and interpret it – depend on the modeler's (your) mindset**

- For (predictive) modelling
  - Target variable (Label) – what we want to predict from the data set
  - All the other variables can be used as predictors
  - Choice of target and predictors depends on your goal
    - Example: Predicting propulsion shaft power using relevant variables available in the *Cruiseships* database

Let's have a look at the data

- In this week's learning material
  - You will find a dataset in csv format (*Cruiseships.csv*). Open the file and take a first look at the data.
  - If any of the variable names are unclear, refer to *CruiseshipsDatabaseAttributeDescription.pdf* for their descriptions.

Let's have a look at the data

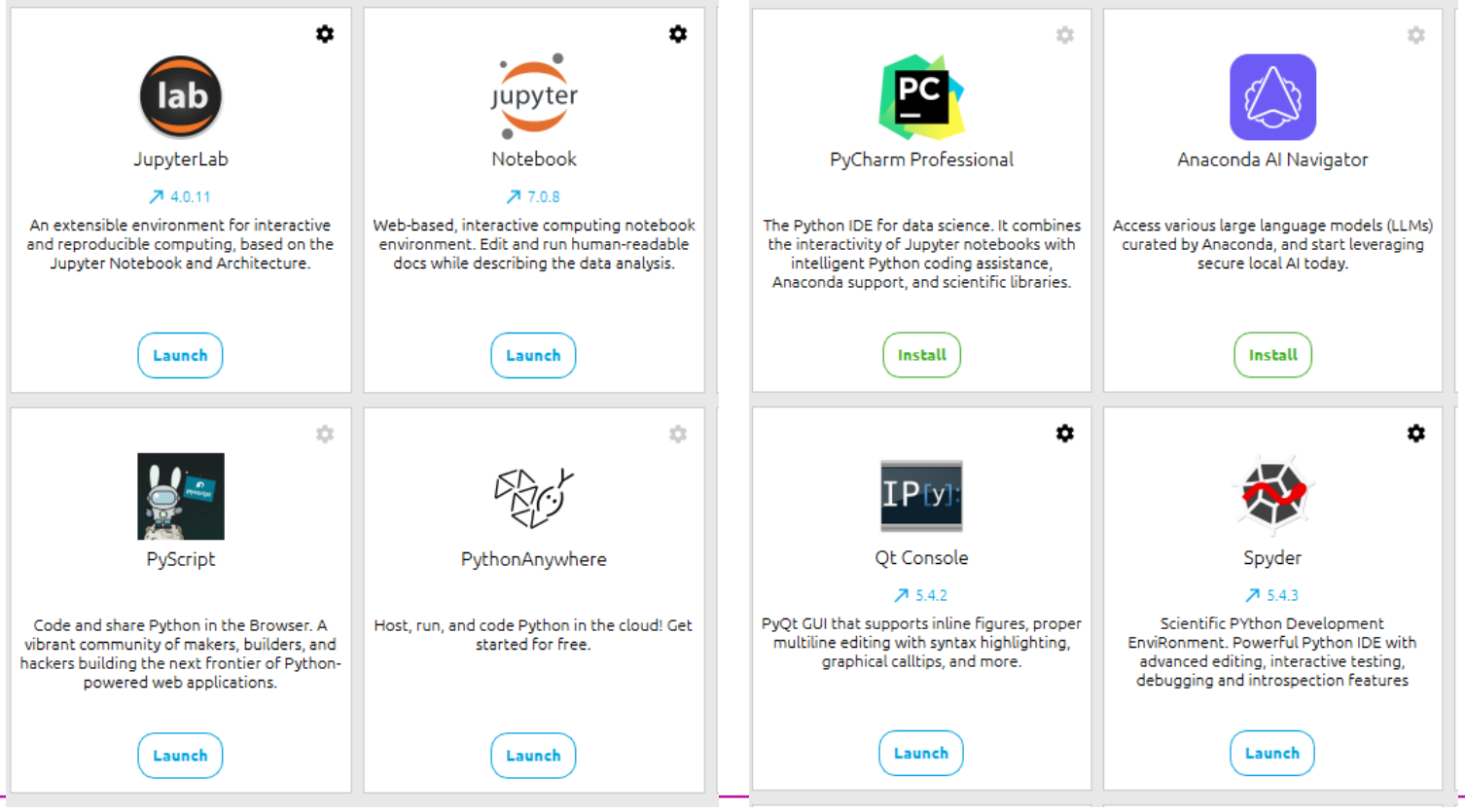
- Variable types
  - Nominal/Categorical: Category, code, state (e.g., name of a ship)
  - Binary: Nominal attribute with only 2 categories or states (e.g., fuel or electric)
  - Ordinal: Attribute with possible values that have a meaningful order or ranking among them (e.g., Class A, Class B, Class C type of vessels)
  - Numeric: A numeric attribute is quantitative; that is, it is a measurable quantity, represented in integer or real values (e.g., ship dimensions)

BREAK (10 minutes)

Programming Language - Python

- **Supports Computational Thinking** – ability to think algorithmically, structure problems, and design reproducible solutions
- Low entry barrier but scalable to advanced tasks
- Transferrable skills across disciplines – statistics, signal processing, AI or numerical modeling …
- Active learning and experimentation
- Accessibility, equity, alignment with open science

Python programming environment



Python programming environment

- **Jupyter notebooks** - computable documents
- Helps in conveying information beyond just the code
- Helpful when information is dynamic/interactive and visually oriented (plots, maps, equations etc.) – particularly important for data analysis
- Works well for small-scale coding
- Easy to get started with when programming experience is limited
- Easy to share

Python programming environment

"We believe that incorporating Jupyter notebooks in our teaching has allowed us to improve 1) engagement, 2) participation, 3) understanding, 4) performance, and 5) preparation for their career"

Barba, L. A., Barker, L. J., Blank, D. S., Brown, J., Downey, A. B., George, T., ... & Zingale, M. (2019). Teaching and learning with Jupyter. Recuperado: https://jupyter4edu. github. io/jupyter-edu-book, 1-77.

Python packages for data analysis

- Python libraries
  - Collection of functions and methods
  - Contain built-in modules providing different functionalities for you to use directly

- We divide the python data analysis libraries into 3 groups

  - Scientific computing libraries
  - Visualization libraries
  - Algorithmic libraries

Python packages for data analysis – Scientific computing libraries







- Pandas (Data structure and tools)
  - Fast access to structured data
  - Primary instrument: data frame (2-dimensional table consisting of column and row labels)
- NumPy (Arrays and matrices)
  - Arrays for inputs and outputs
  - Can be extended to objects
- SciPy (Integrals, solving equations, optimization)
  - Advanced math problems
  - Some data visualization

Python packages for data analysis – Visualization libraries



- Matplotlib (plots and graphs)
  - Most popular
  - Highly customizable

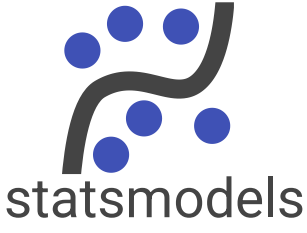


- Seaborn (Complex plots)
  - Based on Matplotlib
  - Heat maps, time series, violin plots

Python packages for data analysis – Algorithmic libraries



- Scikit-learn (Machine learning: regression, classification and more)
  - Most popular for ML
  - Built on NumPy, SciPy, Matplotlib



- Statsmodels (Exploratory analysis, statistical models and tests)
  - Explore data, estimate statistical models, perform statistical tests

Exercise for today: importing and exporting data

Cruiseships.csv <-> MDALecture1ImportExportTemplate.ipynb

Find a peer to work through this exercise

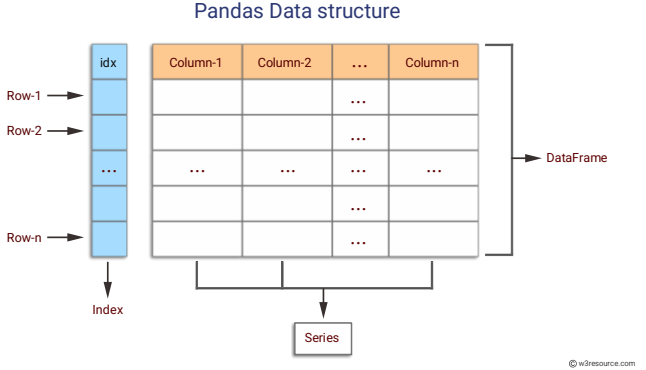
We assume that you have completed the pre-assignment and have the *MDALecture1ImportExportTemplate.ipynb* notebook up and running. If you are not there yet, that is completely fine—please pair with someone who is.

Importing and exporting data using Pandas

- Importing: Process of loading and reading data into Python from various resources

```
Name,Build Year,Passengers (DBL Capacity),Passengers,Crew,L_OA[m],L_PP[m],L[m],Ll
[m],D [m],B [m],T [m],delta SW [ton],Gross Tonnage[ton] ,Deadweight[Ton],C_B,Cb
Alex,Cb Schnee,Cb PNA,Cb ABOUT,LBT,L/delta^(1/3),L/B,B/T,L/T,L/D,P_s [kW],V
[knots],V[m/s],F_n,P/vs^3,Costs[Million],Lyfestyle,Propulsion /
Propellers,Cabins,Ice class,decks(total),decks(passenger),Kolom4,B/D,Number /
Dimensions of the cabins (size range sq m),Cabins outside view / interior or no
view,Kolom5,Kolom6,Kolom7,Kolom8,Kolom9,Kolom10
Majestic
Princess,2017,,3560,1346,330,,,330,,38.4,8.4,,144216,12000,,0.746346065,0.743077807,
0.71215017,0.744711936,106444.8,63.44792348,8.59375,4.571428571,39.28571429,,,,11.3,
0.198603533,0,,,,,,,,,,,,,,,,
Meraviglia,2017,,4500,1536,315.8,,,315.8,,43,8.75,,171598,12200,,0.780278454,0.76327
3821,0.75948387,0.771776138,118819.75,57.29930314,7.344186047,4.914285714,36.0914285
7,,,,9.93,0.178405682,0,,,,,,,,,,,,,,,,
Norwegian
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0.703264061,0.729140123,120396.168,60.97304157,8.055555556,4.747706422,38.24541284,,
,,11.6,0.202803556,0,,,,,,,,,,,,,,,,
```

Function to read the right format of data from the right path and store in a data frame



- What format (.csv, .json, .xlsx, .hdf) does your data come in?
- Where is it located (file path in your computer or online)?

Check: pandas.read_csv()

Importing and exporting data using Pandas

- Importing: Process of loading and reading data into Python from various resources

Step 1: Import the library

To work with data in Python, we use the `pandas` library.

✍️ **Student Input Required:** Import `pandas` and give it the short name `pd`.

```
import pandas as pd
```

Step 2: Load the dataset

We will be using the Cruiseships data available in a CSV (comma separated value) format. The dataset is already made available in your mycourse page.

✅ Hint: Remember, `read_csv()` and `head()` are the methods you need here.

✍️ **Student Input Required:** Read the CSV file into a DataFrame.

```
#df = pd.___('filename.csv')

df = pd.read_csv('C:/Users/musharm1/OneDrive - Aalto University/Teaching/Marine Data Analysis 2025/Datasets/Cruiseships.csv')

# Preview the first 5 rows
df.head(5)
```

Importing and exporting data using Pandas

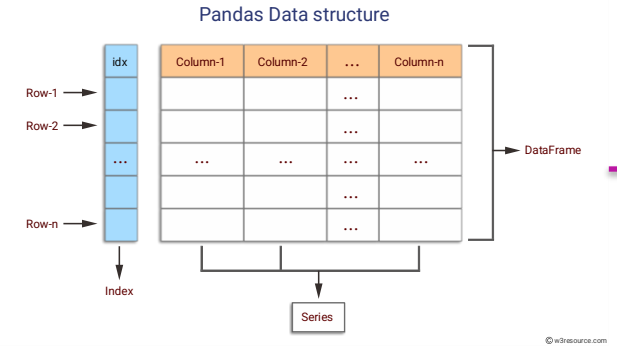
- Print and check the dataframe
- You can print the whole dataframe (can be problematic for large datasets)
- dataframe.head(n) prints the first n rows of the dataframe
- Try dataframe.tail(n) and see what that does

	Name	Build Year	Passengers (DBL Capacity)	Passengers	Crew	L_OA[m]	L_PP[m]	L[m]	LI [m]	D [m]	...	Kolom4	B/D	Number / Dimensions of the cabins (size range sq m)	Cabins outside view / interior or no view	Kolom5	Kolom6	Kolom7
0	Majestic Princess	2017.0	NaN	3560.0	1346.0	330.00	NaN	NaN	330.00	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN
1	Meraviglia	2017.0	NaN	4500.0	1536.0	315.80	NaN	NaN	315.80	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2	Norwegian Joy	2017.0	NaN	3883.0	1700.0	333.50	NaN	NaN	333.50	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN
3	Viking Sky	2017.0	NaN	930.0	NaN	228.20	NaN	NaN	228.20	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN
4	Carnival Vista	2016.0	NaN	3934.0	1450.0	324.00	NaN	NaN	324.00	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN
5	Genting Dream	2016.0	NaN	3352.0	1999.0	335.30	NaN	NaN	335.30	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN
6	Harmony of the seas	2016.0	5479.0	6780.0	2300.0	362.12	NaN	NaN	362.12	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN
7	Koningsdam	2016.0	NaN	2650.0	1036.0	297.00	NaN	NaN	297.00	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN
8	Seven Seas Explorer	2016.0	NaN	750.0	552.0	224.00	NaN	NaN	224.00	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN
9	Antherm of the seas	2015.0	4180.0	4905.0	1500.0	348.00	NaN	NaN	348.00	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN

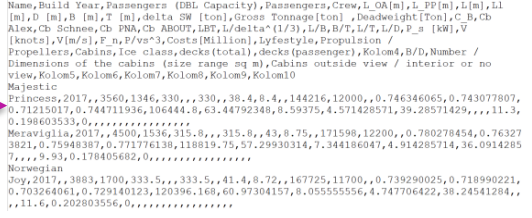
10 rows × 48 columns

Importing and exporting data using Pandas

- Exporting: Process of saving the modified dataset at any stage for future reference



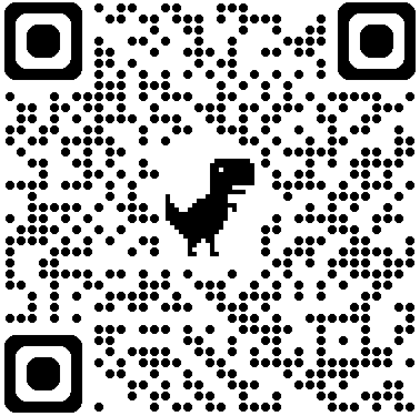
Function to save the data in the right format in the right path.



- What format (.csv, .json, .xlsx, .hdf) do you want to save your data in?
- Where do you want to save the data (file path in your computer)?

Quiz

- Complete your quiz here.



Feedback

- Provide your feedback here

